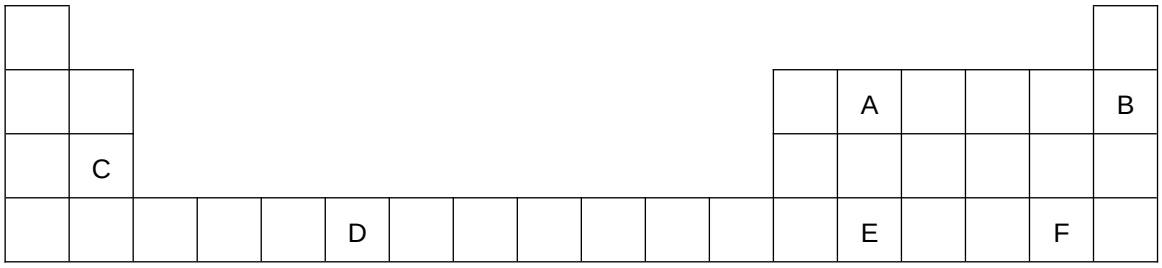
Periodic table

Answers

Classifying the elements

1. Elements can be classified into groups called metals, non-metals and semi-metals (or metalloids). The following diagram illustrates the first four periods of the periodic table. Classify the elements labelled A–F as metals, non-metals or semi-metals.



A: Non-metal

B: Non-metal

C: Metal

D: Metal

E: Semi-metal (metalloid)

F: Non-metal

Family members

2. The following 12 elements belong to four different groups of the periodic table.

potassium, neon, sulfur, fluorine, tellurium, argon, lithium, oxygen, iodine, rubidium, helium, bromine

Sort these elements into their correct groups. In each column, arrange the group members in order of increasing atomic weight.

Lithium	Oxygen	Fluorine	Helium
Potassium	Sulfur	Bromine	Neon
Rubidium	Tellurium	Iodine	Argon

© John Wiley & Sons Australia, Ltd

Something in common

3. For each set of elements, state one characteristic that they have in common.

Set A

(247) Cm Curium 96	(245) Bk Berkelium 97	(251) Cf Californium 98	(254) Es Einsteinium 99	(253) Fm Fermium 100	(256) Md Mendelevium 101

Solid, synthetic, radioactive, metallic

Set B

19.0 F Fluorine 9	20.2 Ne Neon 10
35.5 Cl Chlorine 17	39.9 Ar Argon 18

Gas, non-metal, non-conductors

Set C

58.7 Ni Nickel 28	63.5 Cu Copper 29
106.4 Pd Palladium 46	107.9 Ag Silver 47

Solids, metallic, transition metals, good conductors

Set D

12.0 C Carbon 6	14.0 N Nitrogen 7	16.0 O Oxygen 8

Non-metal

© John Wiley & Sons Australia, Ltd